

Full-Scale Experimental Study on the Effect of CO₂ Flooding on Storage-Seepage Capacity of Tight Sandstone Reservoirs

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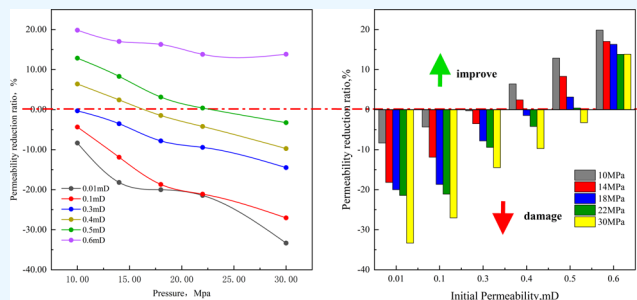
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ABSTRACT: In the CO₂ flooding stage of tight sandstone reservoirs, the CO₂–crude oil–rock interaction causes asphaltene deposition and mineral dissolution effect, which would affect the reservoir performance and reservoir physical property. To reveal the variation law of reservoir performance and percolation capacity in the CO₂ flooding stage, the core samples of typical tight sandstone reservoirs with different permeability levels were selected. An indoor physical flow simulation experiment of CO₂ flooding was then performed. The change in reservoir performance characteristics in the CO₂ flooding stage of the tight sandstone reservoir was quantitatively evaluated, and the evolution law of reservoir macroscopic seepage ability during the CO₂ flooding process was revealed. The experimental results show that the damage degree of CO₂–crude oil–rock interaction on reservoir foundation performance is negatively correlated to the initial permeability of the reservoir. The lower the initial permeability of core samples, the greater would be the damage degree of the experimental process on the effective porosity and permeability of rock samples. However, in reservoirs with relatively high permeability (0.5–0.6 mD), the physical properties of rock samples improved after CO₂ flooding with an average increase in effective porosity and permeability by 4.54 and 10.23%, respectively. The damage degree of reservoir effective porosity is slightly affected by CO₂ injection pressure, while the damage degree of permeability has a significant negative correlation with injection pressure. In the high-pressure miscible flooding stage of 30 MPa, the organic matter deposition caused by the interaction of CO₂–crude oil–rock causes the highest degree of damage to the permeability of rock samples (up to 33.33%). Moreover, the flooding efficiency in the CO₂ flooding stage of a tight sandstone reservoir is positively correlated with the reservoir permeability and CO₂ injection pressure. In the near-miscible flooding stage to miscible flooding stage, the flooding efficiency increases rapidly. After entering the miscible flooding stage, the growth rate in flooding efficiency decreases gradually with the increase in injection pressure.



1. INTRODUCTION

As an unconventional reservoir type, tight sandstone reservoirs play an important role in meeting the growing global energy demand. Tight sandstone reservoirs are characterized by low porosity, low permeability, high content of clay minerals, complex pore throat structure, and strong heterogeneity.^{1–3} CO₂ flooding is an effective way to improve the effect of reservoir development. The effect of the CO₂ flooding mechanism mainly depends on the solubility of CO₂ in crude oil and the interaction between CO₂ and crude oil.⁴ After CO₂ gas is miscible with the crude oil system during the flooding process, it can reduce the viscosity of crude oil, reduce the interfacial tension of the system, and cause the solution gas drive; this is conducive to improving the oil flooding efficiency of tight sandstone reservoirs and improving the reservoir development effect.^{5–10} However, it was found that CO₂ can interact with asphaltene components during CO₂ flooding to cause asphaltene precipitation. The precipitated asphaltene particles migrate with the fluid and block pore throat structures, thereby reducing formation permeability.^{11–17}

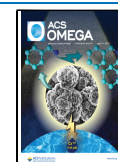
Additionally, after CO₂ dissolved in formation water, it reacts with some minerals. The solid material produced by the reaction blocks the pore throat, decreasing the reservoir permeability and porosity, thereby affecting the recovery of crude oil significantly. This has different degrees of influence on the scale, structure, and seepage capacity of microscopic seepage channels in tight sandstone reservoirs.^{18–21}

CO₂ is commonly used in gas flooding, and its effect in improving oil recovery has been recognized extensively.²² Oomole and Osoba examined the reaction upon putting CO₂, dolomite core, and configured simulated formation water into a reaction vessel.²³ They found that the permeability of the

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core increased significantly after CO₂ injection. After CO₂ interacts with the formation water and rock, the minerals in pores and throats have different degrees of dissolution; therefore, the permeability of the reservoirs increases.²³ Zick first proposed a new flooding type of condensate/evaporation type.²⁴ Zick inferred that this flooding type was not miscible flooding in the strict sense but was characterized by dual functions of condensate and evaporation, low IFT value of oil–gas two-phase, and ≥95% recovery.²⁴ Kamath *et al.* found that compared with the pore throat of conventional reservoirs, the asphaltene deposition produced by CO₂ flooding is more likely to decrease the permeability in tight reservoirs, which is still attributed to the blocking effect of asphaltene deposition on reservoir pore throat.²⁵ Papadimitriou *et al.* found that the effect of asphaltene deposition on larger pore throats (pore size greater than 8 μm) is less than that on smaller pore throats (pore size less than 8 μm), and the space where the smaller pore throat is blocked would reduce the core permeability by 40–90%.²⁶ Wang and Gu experimentally proved that in CO₂ immiscible flooding, increasing the injection pressure would enhance oil recovery; however, because the amount of asphaltene deposition increases significantly with increasing pressure, high-pressure CO₂ flooding would lead to a decrease in reservoir permeability.²⁷ Moreover, experimental data show that even though the injection pressure is higher than minimum miscibility pressure (MMP), the amount of asphaltene deposition continues to increase with the increase in pressure; however, the rate of increase is lower than that before MMP.²⁷ Jafari Behbahani *et al.* experimentally demonstrated that injection pressure has a significant effect on the amount of asphaltene precipitation in the CO₂ flooding process, and asphaltene deposition increases with increasing pressure.²⁸ Deposition of asphaltenes reduces the permeability of the core, thereby reducing CO₂ flooding efficiency and oil recovery.²⁸ Yu *et al.* conducted an indoor CO₂ flooding simulation experiment.²⁹ During the experiment, formation water saturated with CO₂ was injected into the core pores. Following the experiment, they found that the carbonate and silicate minerals were reduced by analyzing the mineral content of the core, and the surface of the two minerals was found to be dissolved using scanning electron microscopy. Dissolution also occurred in some cores containing gypsum.²⁹ Cao and Gu studied the oil recovery degree from the perspective of CO₂ injection pressure and proposed that the oil recovery degree increased significantly near MMP.¹⁵ Additionally, they observed that the amount of asphaltene precipitation increased with the increase in injection pressure, implying that asphaltene precipitation could reduce the ultimate oil recovery by reducing permeability.¹⁵ Wei *et al.* studied the effect of supercritical CO₂ injection on enhanced oil recovery in tight asphaltene formations.³⁰ They focused on the effect of asphaltene precipitation in porous media on rock properties and characterized its changes by using an LF-NMR spectrometer and SEM-EDS. The results show that high injection pressure is beneficial to enhanced oil recovery but accelerates the precipitation of asphaltene in formation. The release of asphaltenes reduces the permeability and porosity of rock significantly, and large pores are more likely to be destroyed.³⁰ Zhao *et al.* injected CO₂ into water-bearing sandstone cores and used nuclear magnetic resonance technology to analyze the T₂ spectrum of sandstone cores before and after CO₂ injection.³¹ The experimental results show that the total volume of small pores increased

significantly, whereas the total volume of medium pores decreased after CO₂ injection. Additionally, after CO₂ injection, the porosity of immobile fluid in small pores increased, whereas the porosity of the immobile fluid in medium pores decreased, and the total porosity, movable fluid porosity, and permeability of tight cores increased significantly.³¹ Su and Yue examined the performance and different mechanisms of CO₂ immiscible flooding over a permeability range between 0.4 and 324 mD by core flood experiments and oil–gas interaction tests.³² Different mechanisms of CO₂ immiscible flooding, including gas solubility, oil viscosity, CO₂–oil IFT, CO₂ extraction, and oil swelling were analyzed in detail. Moreover, the capillary pressure and the characteristic structure coefficient were used for further interpretation. The results show that the CO₂ immiscible flooding efficiency inversely varies with permeability, but the trend of it is significantly different before and after a threshold permeability K-c.³² Cai *et al.* used online low-field nuclear magnetic resonance (NMR) imaging to monitor CO₂ flooding under different reservoir pressures in low permeability siltstone core plugs presaturated with stock tank oil from the Chang7 Formation in China.³³ To visualize the oil distribution, NMR T₂ and T₁–T₂ spectra, one-dimensional frequency (1D-FQ), and magnetic resonance imaging (MRI) were conducted to obtain the overall and spatially distributed saturations. The results show that recovery by miscible flooding (69.4%) was twice more than that by immiscible flooding (32.6%). The CO₂ retention rate following miscible flooding was 10.59% higher than that due to immiscible flooding, primarily due to remnant dissolved CO₂ in the oil. In contrast, immiscible flooding limited the solubility of CO₂ and led to solubility and residual trapping.³³ Zhao and Yin conducted experiments under pressure boost and pressure reduction conditions to simulate CO₂–brine scaling in different parts of the reservoir during CO₂ flooding and experiments on the dissolution and scaling of CO₂–brine–rock were carried out.³⁴ The results show that the increase of permeability of core saturated high scale-forming ions is smaller than that of saturated no scale-forming ions brine after CO₂ flooding. The accumulation and attachment of salt crystals were found in some large pores of the core with scale-forming ion water after CO₂ flooding. The ratio of medium-sized pores decreased, while that of large and small pores increased, and the pore radius distribution differentiated toward polarization.³⁴ Li *et al.* carried out a series of diagenetic reaction simulation experiments in a semiclosed high-temperature and high-pressure simulation system by using potassium feldspar samples, albite samples, and calcite samples with CO₂ reaction solution.³⁵ The experimental results show that under the conditions of high temperature and high pressure, CO₂-rich fluid has obvious dissolution effect on potassium feldspar, albite, and calcite. The pressure changes the solubility of CO₂, which affects the dissolution effect and strength of sandstone components.³⁵

Existing studies evaluate the influence of the asphaltene deposition effect and mineral dissolution effect on the reservoir in the CO₂ flooding stage separately but do not combine the two effectively and conduct experimental research at the same scale. Moreover, studies on the damage assessment of CO₂–crude oil–rock interaction on tight sandstone reservoirs only focus on individual permeability reservoirs. However, systematic and comprehensive damage assessment studies on tight sandstone reservoirs with different levels of permeability

Table 1. Core Physical Parameters

level of permeability	sample ID	length/cm	diameter/cm	porosity/%	permeability/ $10^{-3} \mu\text{m}^2$	pressure/MPa	displacement state
0.01	1-1	9.85	2.50	5.623	0.012	10.00	immiscible
	1-2	8.95	2.49	5.615	0.011	14.00	near-miscible
	1-3	8.59	2.51	4.327	0.010	18.00	miscible
	1-4	8.75	2.50	5.695	0.014	22.00	miscible
	1-5	9.65	2.49	4.032	0.012	30.00	miscible
0.1	2-1	8.65	2.51	7.132	0.115	10.00	immiscible
	2-2	9.65	2.50	6.959	0.101	14.00	near-miscible
	2-3	9.74	2.50	7.212	0.091	18.00	miscible
	2-4	8.94	2.51	5.695	0.090	22.00	miscible
	2-5	10.21	2.51	6.276	0.111	30.00	miscible
0.3	3-1	9.69	2.51	10.613	0.325	10.00	immiscible
	3-2	9.66	2.50	9.656	0.312	14.00	near-miscible
	3-3	9.47	2.51	8.935	0.307	18.00	miscible
	3-4	10.54	2.51	9.242	0.297	22.00	miscible
	3-5	10.14	2.49	8.989	0.290	30.00	miscible
0.4	4-1	7.85	2.49	10.525	0.422	10.00	immiscible
	4-2	8.99	2.50	12.241	0.414	14.00	near-miscible
	4-3	9.03	2.49	11.256	0.411	18.00	miscible
	4-4	9.65	2.51	9.569	0.409	26.00	miscible
	4-5	8.98	2.50	11.254	0.391	30.00	miscible
0.5	5-1	10.23	2.49	11.301	0.521	10.00	immiscible
	5-2	11.38	2.50	12.365	0.518	14.00	near-miscible
	5-3	9.65	2.51	11.256	0.509	18.00	miscible
	5-4	7.86	2.49	9.898	0.495	22.00	miscible
	5-5	8.36	2.51	10.569	0.491	30.00	miscible
0.6	6-1	9.45	2.49	12.236	0.615	10.00	immiscible
	6-2	9.74	2.50	11.365	0.611	14.00	near-miscible
	6-3	9.14	2.51	10.325	0.626	18.00	miscible
	6-4	8.25	2.49	12.369	0.601	22.00	miscible
	6-5	10.14	2.51	11.367	0.578	30.00	miscible

(covering high, medium, and low permeability) have not been conducted.

Therefore, to reveal the variation characteristics of reservoir storage and seepage capacity during CO₂ flooding in tight sandstone reservoirs and to realize the full coverage evaluation of the evolution of tight sandstone reservoir performance in the CO₂ flooding stage, we selected typical tight sandstone core samples with different permeability levels to perform the CO₂ flooding physical simulation experiment to quantitatively evaluate the change in reservoir performance characteristics in the CO₂ flooding stage of the tight sandstone reservoir, reveal the evolution law of macroscopic storage capacity and seepage capacity in the process of CO₂ flooding, and screen the main controlling factors to provide a theoretical basis for optimizing the CO₂ flooding injection and production plan to realize the efficient development of tight sandstone reservoirs.

2. EXPERIMENTAL SECTION

2.1. Materials. In this experiment, the tight sandstone core of the Chang 8 reservoir in the Ordos Basin was selected to conduct the physical flow simulation experiment of CO₂ flooding. The core was divided into six groups, according to different permeability levels, with five cores in each group and 30 samples in total. The core samples are from the tight sandstone reservoir of the Chang 8 member of Triassic Yanchang Formation in the Ordos Basin, and the coring depth range is 2145.25–2182.38 m. The core samples in the sampling area are mainly gray and dark gray fine-grained feldspar quartz sandstone and feldspar lithic sandstone, and

there are some dark gray and dark mudstone interbeds. The experimental results of X-ray diffraction analysis of the selected core show that the quartz content is the highest in all minerals, with an average of 44.05%. Followed by plagioclase, the average mineral content is 21.01%, the feldspar content is 8.40%, and the micro plagioclase content is 4.16%. The clay content is 10.48–14.69%, the average carbonate mineral content is 4.66%, and the pyrite content is 1.79%. In the process of experiment, the crude oil of the Chang 8 reservoir in the coring block was selected as the experimental oil sample. The density of the oil sample was 775 kg/m³, the viscosity was 1.68 mPa·s under the pressure of 14 MPa at the reservoir temperature, the asphaltene content in the crude oil sample was 1.65 wt %, and the resin content was 5.72 wt %. The specific size, physical properties, and experimental conditions are shown in Table 1. As shown in Figure 1, the minimum miscible pressure in the study area is 15.8 MPa determined by the slim tube experiment.

2.2. Experimental Devices. Figure 2 shows the experimental equipment, which is composed of the displacement system, core holder, back pressure system, and calculation system. The displacement pump used in the displacement system is type 260D with two pump bodies, produced by Isco Company in the United States. The volume of the pump body was 103.0 mL, the minimum flow rate could be set as 0.001 mL/min, the maximum flow rate could be set as 80.000 mL/min, and the maximum displacement pressure was 51.7 MPa.

2.3. Experimental Procedures.

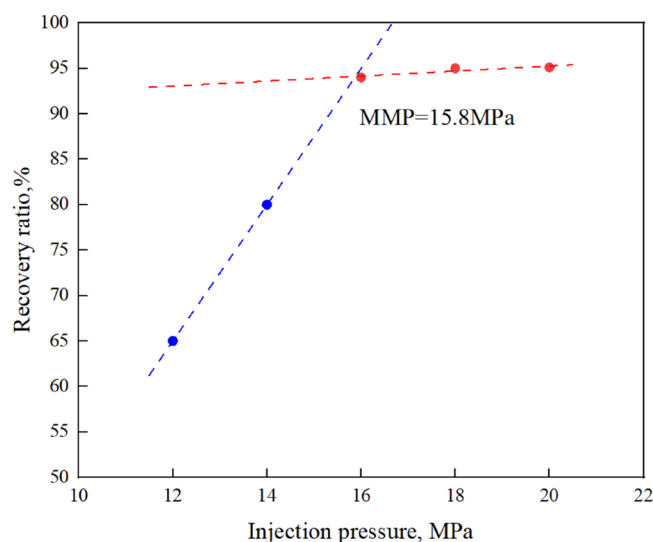


Figure 1. Minimum miscible pressure determination curve.

- (i) Core samples were screened according to the permeability level. The cores were washed with petroleum ether and benzene at 5 MPa and 85 °C for 10 days and then dried at 90 °C for 12 h.
- (ii) The porosity and gas permeability parameters of the sample were tested, and the sample was dried after the test. The core porosity measurement was performed by the liquid saturation method, and the permeability of the core samples was measured by the steady-state method using a permeability tester (Huaxing Co., Nantong, China). With this method, nitrogen was injected into the core sample with a constant injection rate. Then, by measuring pressure difference across the whole core sample, the gas permeability of the core sample can be obtained from Darcy's law.
- (iii) The simulated formation water was configured, and the vacuum pump was used to saturate the core sample for 48 h.

- (iv) The Mn^{2+} solution with a mass concentration of 15,000 mg/L was injected into the core to displace the simulated formation water at a constant speed of 0.05 mL/min. The injection volume was 3–4 pore volume (PV).
- (v) The experimental crude oil sample was injected into the core at a constant rate of 0.05 mL/min until the oil content of the produced fluid at the core outlet reached 100%, and the oil–water distribution model of the original formation was established.
- (vi) The back pressure valve was controlled to stabilize the CO_2 injection pressure. The core samples were displaced at a constant pressure of 10.0, 14.0, 18.0, 22.0, and 30.0 MPa until no crude oil flowed outside the outlet. The confining pressure was set to 35 MPa.
- (vii) The inlet and outlet ends were closed, aging the core for 120 h to fully react with the CO_2 -rich fluid with the core matrix minerals to simulate the CO_2 flooding reservoir dissolution effect.
- (viii) The porosity and gas permeability parameters of the core samples after the experiment were tested again, and the changes in core seepage capacity were evaluated by comparing the parameters of the first test samples.

3. RESULTS AND DISCUSSION

In this section, the samples with initial permeabilities of 0.01 and 0.1 mD were designated as low-permeability reservoirs, samples with initial permeabilities of 0.3 and 0.4 mD were designated as medium permeability reservoirs, and samples with initial permeabilities of 0.5 and 0.6 mD were designated as high-permeability reservoirs. The minimum miscible pressure in the study area is 15.8 MPa determined by the slim tube experiment. Based on reservoir porosity and permeability variation before and after the experiment, the injection pressure is set as the abscissa and the porosity/permeability reduction ratio is set as the ordinate to evaluate the change law of porosity and permeability of core samples with different permeability levels before and after CO_2 flooding. The changes in macroscopic physical properties of reservoirs under different

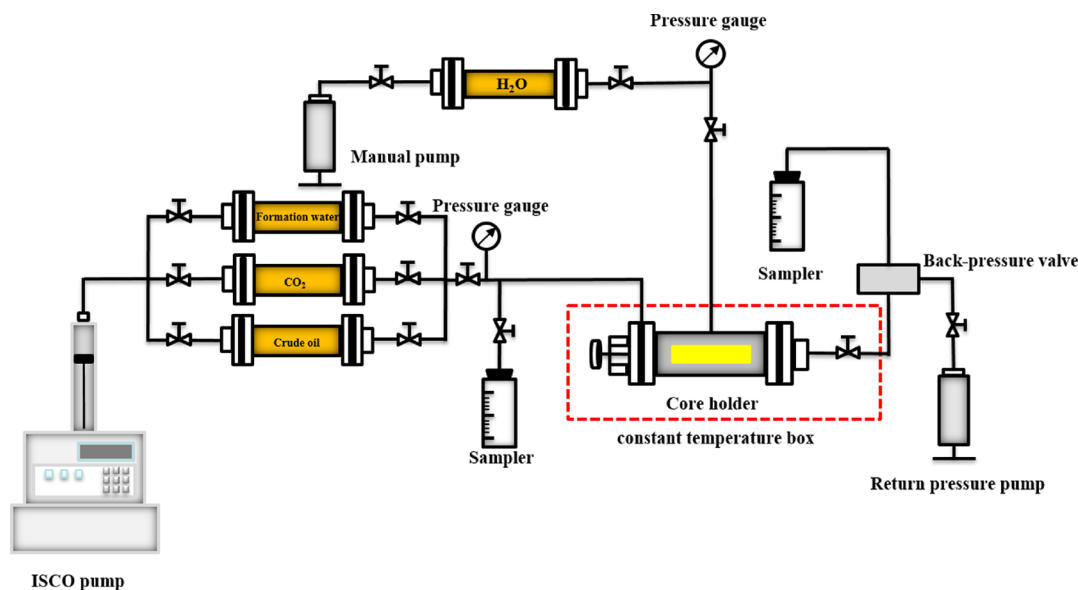
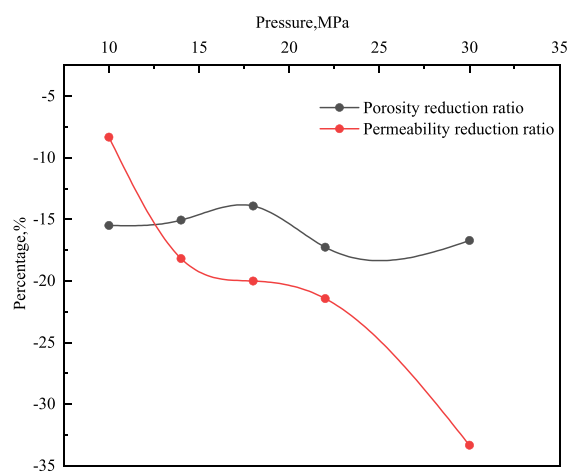


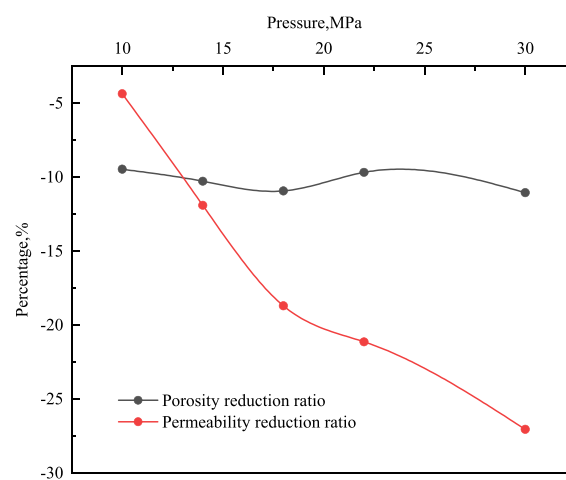
Figure 2. Schematic diagram of the CO_2 flooding device.

Table 2. Core Physical Damage Rate

sample ID	displacement state	pressure/MPa	permeability/ $10^{-3} \mu\text{m}^2$	porosity damage rate/%	permeability damage rate/%
1-1	immiscible	10	0.012	−15.49	−8.33
1-2	near-miscible	14	0.011	−15.07	−18.18
1-3	miscible	18	0.01	−13.91	−20.00
1-4	miscible	22	0.014	−17.26	−21.43
1-5	miscible	30	0.012	−16.72	−33.33
2-1	immiscible	10	0.115	−9.45	−4.35
2-2	near-miscible	14	0.101	−10.26	−11.88
2-3	miscible	18	0.091	−10.91	−18.68
2-4	miscible	22	0.09	−9.66	−21.11
2-5	miscible	30	0.111	−11.03	−27.03
3-1	immiscible	10	0.325	−9.70	−0.31
3-2	near-miscible	14	0.312	−7.88	−3.53
3-3	miscible	18	0.307	−8.88	−7.82
3-4	miscible	22	0.297	−8.84	−9.43
3-5	miscible	30	0.29	−9.62	−14.48
4-1	immiscible	10	0.422	−4.03	6.40
4-2	near-miscible	14	0.414	−3.12	2.42
4-3	miscible	18	0.411	−2.96	−1.46
4-4	miscible	22	0.405	−2.99	−4.20
4-5	miscible	30	0.391	−3.72	−9.72
5-1	immiscible	10	0.521	1.72	12.86
5-2	near-miscible	14	0.518	0.56	8.30
5-3	miscible	18	0.509	0.97	3.14
5-4	miscible	22	0.495	3.57	0.40
5-5	miscible	30	0.491	2.72	−3.26
6-1	immiscible	10	0.615	6.12	19.84
6-2	near-miscible	14	0.611	5.81	17.02
6-3	miscible	18	0.626	10.07	16.29
6-4	miscible	22	0.601	7.24	13.81
6-5	miscible	30	0.578	6.67	13.84



(a) Reduction ratio of porosity and permeability at 0.01 mD



(b) Reduction ratio of porosity and permeability at 0.1 mD

Figure 3. Reduction ratio of porosity and permeability in the low permeability group: (a) 0.01 mD and (b) 0.1 mD.

permeability levels were compared to reveal the variation characteristics of the macroscopic seepage capacity of reservoirs during CO_2 flooding. The porosity and permeability damage rate of the core after the experiment are shown in Table 2.

3.1. Evaluation of Macroscopic Seepage Capacity Based on the Reservoir Permeability Level. Cores 1-1 to 1-5 indicate the group with the lowest permeability (0.01 mD) in this experiment. As shown in Figure 3a, the average permeability reduction ratio of this sample group is −20.26%, and the average porosity reduction ratio is −15.69%. The

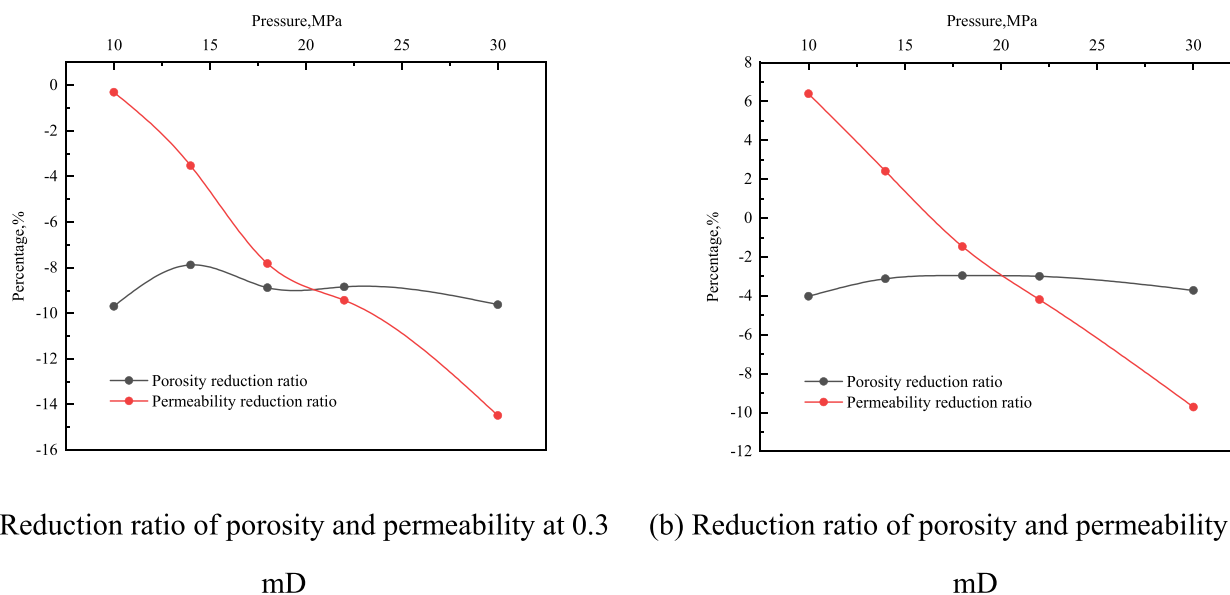


Figure 4. Reduction ratio of porosity and permeability in the medium permeability group: (a) 0.3 mD and (b) 0.4 mD.

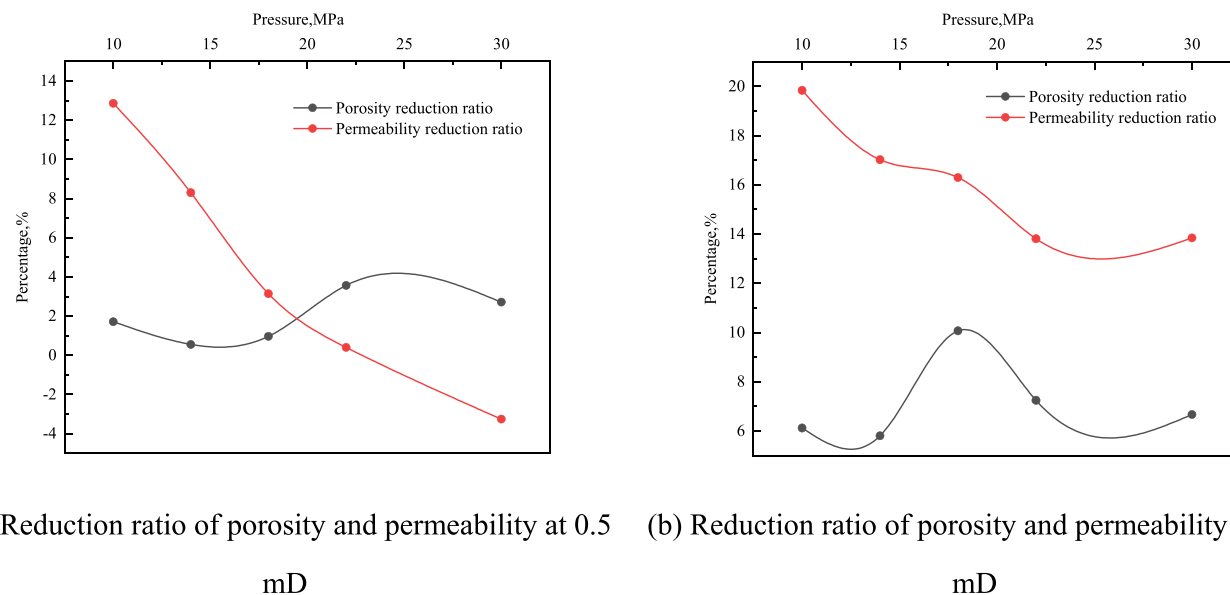


Figure 5. Reduction ratio of porosity and permeability in high permeability group (a) 0.5 mD and (b) 0.6 mD.

permeability damage rate of core 1-5 is the highest at -33.33% , and the porosity damage rate of core 1-4 is the highest at -17.26% . Cores 2-1 to 2-5 had an initial average permeability of approximately 0.1 mD with a maximum of 0.115 mD and a minimum of 0.09 mD. As shown in Figure 3b, the average reduction ratios of permeability and porosity in the experiment are -16.61 and -10.26% , respectively. Among them, the permeability damage rate of core 2-5 approximately -27.03% , whereas the porosity damage rate is -11.03% .

As shown in Figure 4a, the average reduction ratio of reservoir permeability is -7.11% and the average reduction ratio of porosity is -8.98% when the initial permeability is in the 0.3 mD sample group. The permeability damage rate of core 3-5 is up to -14.48% , and the porosity damage rate of core 3-1 is up to -9.70% . Cores 4-1 to 4-5 had an initial average permeability of 0.4 mD. After the experiment, the average reduction ratios of permeability and porosity were -1.31 and -3.36% , respectively. As shown in Figure 4b, the

permeability change value of the sample group with a permeability of 0.4 mD was positive in the experimental stage, and the reservoir permeability increased for certain injection pressures.

As shown in Figure 5a, the initial average permeability of cores 5-1 to 5-5 was 0.5 mD. This sample group showed significant permeability improvement characteristics with an average of 4.29%. Porosity increased gradually with an average rate of change of 1.90%. At this time, core 5-1 had the largest permeability increase (12.86%), and core 5-4 had the highest porosity improvement (3.57%). The initial average permeability of cores 6-1 to 6-5 reached 0.6 mD, which is the sample group with the highest permeability in this experiment. The porosity and permeability of the samples improved throughout the process with an average porosity increase of 7.18% and average permeability increase of 16.16%. Core 6-1 had the largest permeability increase (19.84%), and core 6-3 had the largest porosity increase (10.07%) (Figure 5b).

Comprehensive analysis of Figures 6 and 7 shows that the effective porosity and permeability of low-permeability

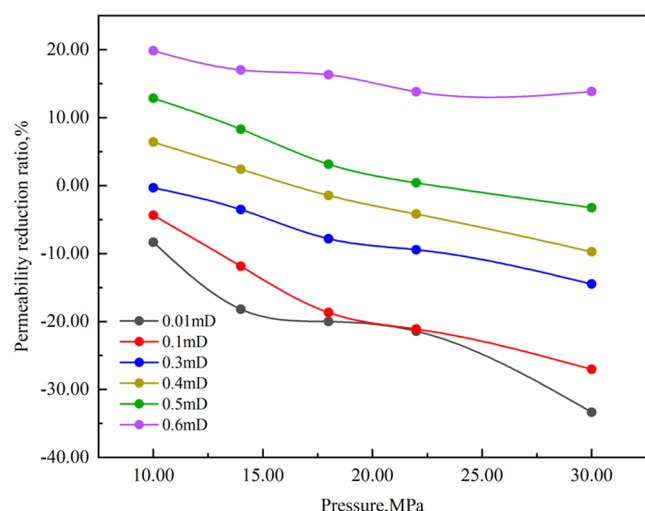


Figure 6. Relationship curve between permeability reduction ratio and pressure for each permeability level.

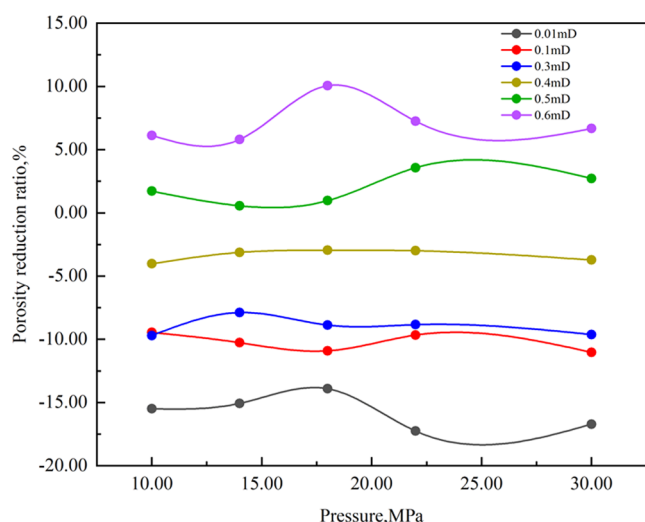


Figure 7. Relationship curve between porosity reduction ratio and pressure for each permeability level.

reservoirs (0.01–0.1 mD) and medium permeability reservoirs (0.3–0.4 mD) reduced significantly after CO₂ flooding. The flooding process has different degrees of damage effects on the macrophysical properties of cores. With the increase in permeability, the change rate of core porosity and permeability became positive after CO₂ flooding, indicating that the core physical properties of high-permeability reservoirs (0.5–0.6 mD) improved to a certain extent. Owing to the large scale of the seepage channel, the organic sediment can be flushed out with the fluid. Mineral dissolution occurs in the CO₂ flooding process, which expands the reservoir seepage channel and improves the core physical property.

The experimental results show that the damage degree of core physical properties is negatively correlated with initial permeability. Since the absolute content of asphaltene deposition and mineral dissolution is small relative to pore volume, the degree of pore failure is less affected by injection pressure. The damage rate of porosity in low-permeability

cores is generally higher and decreases with the increase in initial permeability of the core. The change in permeability is similar to the change in porosity; however, the change in permeability is negatively correlated with the injection pressure. It is evident from Figure 5a,b that with the increase in injection pressure, the improvement in core permeability decreases gradually or even turns into damage. We concluded that when the injection pressure is high, the deposition of organic matter is large, and the size of precipitated particles increases gradually with the migration of fluid in the rock seepage channel. When the particle size of organic matter deposited approaches or exceeds the size of the pore throat, the particles would be captured, thereby blocking the narrow throat and reducing the pore throat connectivity of the rock. Simultaneously, organic matter deposited is adsorbed continuously on the surface of rock minerals, thereby reducing the size of the seepage channel gradually, making it easier for organic matter deposition particles to plug the pore throat.

3.2. Macroscopic Seepage Capacity Evaluation Based on Flooding Phase States. As the minimum miscibility pressure between the formation crude oil samples and CO₂ measured by the slim tube experiment is 15.8 MPa, the injection pressure of 10 MPa is set as the immiscible flooding stage, and the injection pressure of 14 MPa is set as the near-miscible flooding stage. When the injection pressure (18, 22, and 30 MPa) is greater than the minimum miscible pressure, it is set as the miscible flooding stage, and 30 MPa is the high-pressure miscible flooding stage. Based on reservoir porosity and permeability variation before and after the experiment, the initial permeability of the core was set as the abscissa and porosity/permeability reduction ratio was set as the ordinate. To evaluate the changes in porosity and permeability of core samples in different flooding phases before and after CO₂ flooding, changes in macroscopic physical properties of reservoirs, under the condition of injection pressure difference, were fully compared, and the changes in macroscopic seepage capacity of reservoirs in the CO₂ flooding phase were revealed.

Figure 8 shows the relationship between initial core permeability and permeability variation under different injection pressures. In the immiscible phase (10 MPa), with

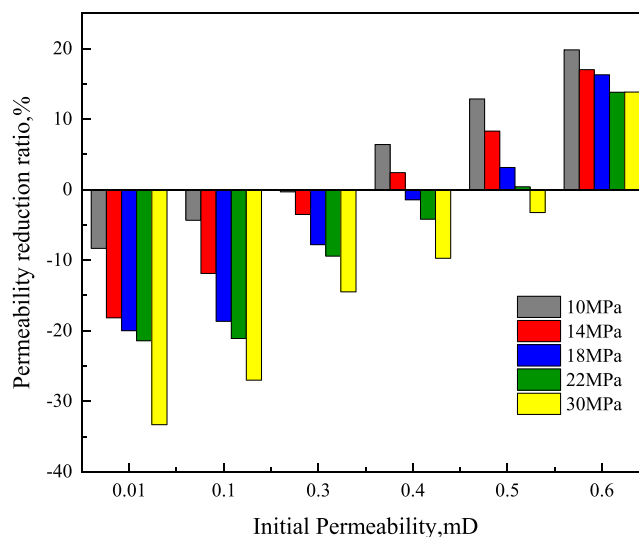


Figure 8. Relationship between initial permeability and permeability reduction ratio under different injection pressures.

the increase in initial permeability, the absolute value of the core permeability reduction ratio decreases, and the damage degree decreases gradually. When the initial permeability of the core is greater than 0.3 mD, the core permeability improves, and the permeability rise ratio increases with the increase in the initial permeability of the core. In the near-miscible phase (14 MPa), with the increase in initial permeability, the absolute value of the reduction ratio of core permeability decreases, and the damage degree decreases gradually. When the initial permeability of the core is greater than 0.4 mD, the core permeability improves, and the permeability rise ratio increases with the increase in the initial permeability of the core. When the injection pressure is greater than the minimum miscibility pressure, the deposition of organic matter continues to increase, and the damage to core permeability increases. By comparing the permeability reduction ratio of each core for different injection pressures, we observed that the permeability damage rates of the three cores with an initial permeability of 0.01 mD were the highest in the same group and increased with the increase in injection pressure. With the increase in injection pressure, the permeability damage degree of core samples with the same initial permeability level increases, while the improvement degree of core samples with high permeability decreases. The reduction ratio of core permeability is negatively correlated with injection pressure. From the immiscible phase to the high-pressure miscible phase, the permeability damage degree of low-permeability cores is much higher than that of high-permeability cores, and the permeability reduction ratio of the 0.01 mD core decreases by 25.00% from -8.33 to -33.33% . However, the permeability reduction ratio of 0.6 mD core samples decreased from 19.84 to 13.84% (i.e., only a 6.00% decrease) (Table 2). Simultaneously, the critical point between damage and improvement of core permeability gradually moves backward with the rise in injection pressure. In the miscible phase, the critical point only appears in the core samples with high permeability.

Figure 9 shows the relationship between initial porosity and permeability reduction ratio under different injection pressures. The figure indicates that compared with the permeability

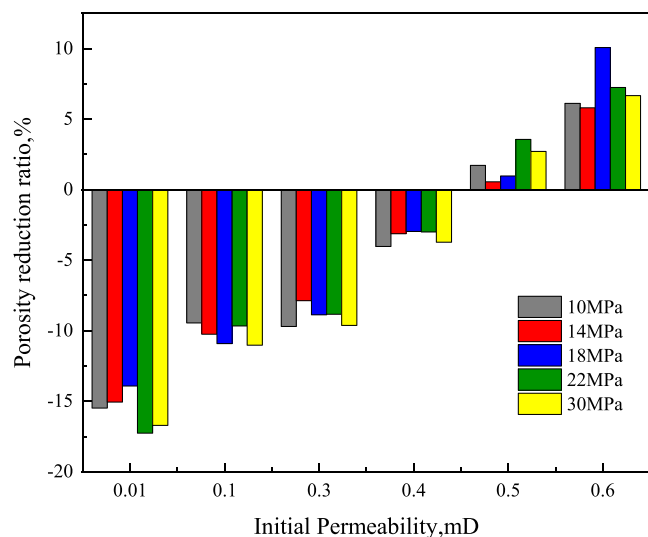


Figure 9. Relationship between initial permeability and reduction ratio of porosity under different injection pressures.

reduction ratio, the relationship between the reduction ratio of effective porosity and injection pressure (phase) is not evident. With the increase in initial core permeability, the porosity reduction ratio changes from damage to improvement, and the improvement phenomenon appears only in the high-permeability core experiment. In the case of the same asphaltene deposition rate, the plugging rate of pore throat is inversely proportional to the core permeability and pore throat size. The asphaltene deposition attached to the pore wall will reduce the pore space, while the asphaltene deposition deposited in the throat and pore stenosis will form blockage. However, if the core permeability and pore throat scale are large, the asphaltene deposition attached to the pore throat wall will not significantly reduce the pore space of the reservoir, and the sediments accumulated in the throat and narrow place also have difficulty forming effective blockage of the pore throat. Due to the large scale of the seepage channel, the organic sediment is flushed out with the fluid, which enlarges the reservoir seepage channel and improves the core physical property. The porosity damage degree decreases with the increase in initial core permeability under different flooding phases, and the highest porosity damage degree is at 0.01 mD.

In the immiscible phase, the amount of organic matter deposited is small, and the permeability damage degree of the reservoir is low. The initial permeability level of 0.4 mD is the critical point from permeability damage to improvement. With the increase in injection pressure, CO_2 continues to dissolve in the crude oil, resulting in an increase in the deposition of organic matter, leading to the blockage of seepage channels and increase in core permeability damage. At this time, the critical point of permeability damage to improve is pushed back to 0.6 mD, that is, the damage caused by organic matter deposition needs a larger pore throat scale to offset. The reduction ratio of porosity follows the same law as permeability; however, the reduction ratio of porosity is less sensitive to displacement phase states compared with permeability.

3.3. Evaluation of Change Law of Flooding Efficiency.

Figure 10 shows the relationship between injection pressure and oil flooding efficiency of samples with different

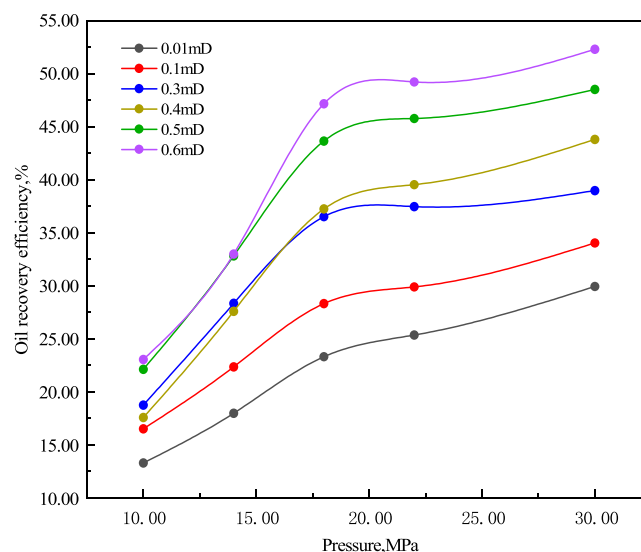


Figure 10. Relationship between pressure and oil flooding efficiency of different permeability levels.

permeability levels. The experimental results show that there is a positive correlation between oil flooding efficiency and injection pressure in the CO₂ flooding stage, that is, the higher the injection pressure, the higher would be the oil flooding efficiency. The first stage is from near-miscible flooding to miscible flooding. In this stage, a large amount of CO₂ is dissolved in crude oil, which gives full play to the advantages of reducing interfacial tension, increasing elastic energy, and reducing crude oil viscosity, and the rapid improvement of oil flooding efficiency is realized. However, the growth rate of oil flooding efficiency of core samples with different levels of permeability is different. The higher the permeability, the greater would be the oil flooding efficiency. After CO₂ flooding enters miscible flooding, the oil flooding efficiency continues to rise with the increase in pressure. However, the oil flooding efficiency curves of the six groups of samples all enter the plateau stage, indicating that the oil flooding efficiency growth rate at this time is significantly lower than that from near-miscible flooding to the miscible flooding stage. Finally, under the maximum injection pressure of 30 MPa, the oil flooding efficiency reached the highest value. The final recovery efficiency of 0.60 mD samples reached 52.31%, which was 22.38% higher than that of 0.01 mD samples (Table 3). As can be seen from Figure 10, the oil production efficiency curve of the 0.3, 0.5, and 0.6 mD groups entered the plateau period between the pressures of 18 and 20 MPa, and the current injection pressure is the optimal injection pressure.

Table 3. Core Sample Flooding Efficiency

sample ID	displacement state	pressure/MPa	permeability/ $10^{-3} \mu\text{m}^2$	oil flooding efficiency/%
1-1	immiscible	10	0.012	13.28
1-2	near-miscible	14	0.011	17.97
1-3	miscible	18	0.01	23.31
1-4	miscible	22	0.014	25.35
1-5	miscible	30	0.012	29.93
2-1	immiscible	10	0.115	16.5
2-2	near-miscible	14	0.101	22.35
2-3	miscible	18	0.091	28.32
2-4	miscible	22	0.09	29.89
2-5	miscible	30	0.111	34.04
3-1	immiscible	10	0.325	18.74
3-2	near-miscible	14	0.312	28.34
3-3	miscible	18	0.307	36.52
3-4	miscible	22	0.297	37.46
3-5	miscible	30	0.29	38.98
4-1	immiscible	10	0.422	17.56
4-2	near-miscible	14	0.414	27.59
4-3	miscible	18	0.411	37.25
4-4	miscible	22	0.405	39.54
4-5	miscible	30	0.391	43.81
5-1	immiscible	10	0.521	22.12
5-2	near-miscible	14	0.518	32.82
5-3	miscible	18	0.509	43.65
5-4	miscible	22	0.495	45.78
5-5	miscible	30	0.491	48.52
6-1	immiscible	10	0.615	23.04
6-2	near-miscible	14	0.611	32.99
6-3	miscible	18	0.626	47.18
6-4	miscible	22	0.601	49.23
6-5	miscible	30	0.578	52.31

Figure 11 shows the relationship between permeability and oil flooding efficiency of six groups of experimental samples

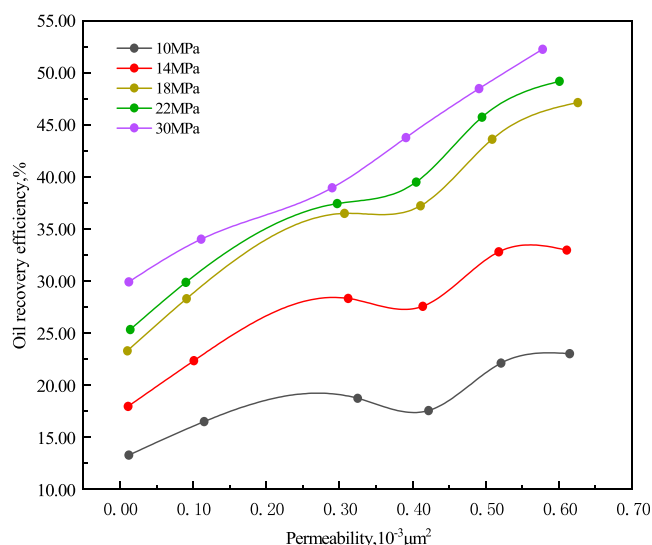


Figure 11. Relationship between permeability and oil flooding efficiency.

under different injection pressures. The experimental results show that under the same injection pressure, the oil flooding efficiency of the sample is positively correlated with the initial permeability, that is, the higher the sample permeability, the higher would be the oil flooding efficiency. However, it can be observed from Figure 11 that when the injection pressure increases gradually, the flooding efficiency also increases gradually. The injection pressures in the experimental process are set to be 10.0, 14.0, 18.0, 22.0, and 30.0 MPa. The gray curve shows the immiscible flooding stage at 10.0 MPa injection pressure. As the sample permeability increases from 0.01 to 0.6 mD, the oil flooding efficiency shows a slow upward trend (13.28 to 23.04%). When the experimental pressure increases gradually to 14.0 MPa and enters the near-miscible flooding stage, the red curve is higher than the gray curve. With the increase in the initial permeability of the sample, the oil flooding efficiency increases from 17.97 to 32.99% (Table 3), and the oil flooding efficiency at each point is higher than that at the immiscible flooding stage. When the experimental pressure continues to rise to 18.0 MPa and enters the miscible flooding stage, the yellow curve of the overall oil flooding efficiency is much higher than the gray and red curves of the immiscible stage, and the increase of oil flooding efficiency reaches the maximum. With the increase in initial core permeability, the oil flooding efficiency increases from 23.31 to 47.18% (Table 3), and the overall oil flooding efficiency increases by 9.02% on average compared with the 14 MPa flooding experiment. After entering the miscible phase, with the rise of injection pressure, the overall oil flooding efficiency still showed a continuous upward trend; however, the increase in flooding efficiency curves of each injection pressure decreased significantly. The flooding efficiency curves of 18 to 30 MPa converged with a small difference in magnitude. Finally, for the maximum injection pressure of 30.0 MPa, the oil flooding efficiency reaches the highest value of 52.31% (Table 3). Meanwhile, the oil flooding efficiency of the sample with 0.01 mD permeability is much higher than that of the sample with 0.6 mD permeability in the immiscible flooding

stage and even close to that of the sample with 0.6 mD permeability in the near-miscible flooding stage.

The comprehensive analysis shows that there is a positive correlation between oil flooding efficiency and injection pressure in the CO₂ flooding stage, and the change in injection pressure has a significant impact on oil flooding efficiency. From near-miscible flooding to miscible flooding stage, the oil flooding efficiency increases rapidly. After entering the miscible flooding stage, the increase in oil flooding efficiency gradually decreases with the increase in injection pressure. The permeability of core samples is another important factor affecting CO₂ flooding efficiency. Further, the permeability is positively correlated with oil flooding efficiency. The better the reservoir physical property and percolation ability, the higher would be the CO₂ flooding efficiency.

4. CONCLUSIONS

In this study, the typical tight sandstone core samples with different permeability levels were selected to perform the physical simulation experiment of CO₂ flooding. The variation characteristics of reservoir performance during CO₂ flooding in tight sandstone reservoirs were quantitatively evaluated, and the evolution law of macroscopic reservoir and seepage capacity during CO₂ flooding was revealed. The detailed conclusions are summarized as follows:

- (i) The damage degree of CO₂–crude oil–rock interaction on reservoir basic properties is negatively correlated with the initial permeability of the reservoir, that is, the lower the initial permeability of core samples, the greater would be the damage degree of the experimental process on the effective porosity and permeability of rock samples.
- (ii) The effective porosity damage degree of reservoir is less affected by the injection pressure, while the permeability damage degree has a significant negative correlation with the injection pressure. With the increase in injection pressure (from immiscible stage to miscible stage), the permeability damage degree of rock samples increases gradually.
- (iii) The initial reservoir permeability in the CO₂ flooding stage of a tight sandstone reservoir directly affects the oil flooding efficiency. There is a significant positive correlation between reservoir permeability and oil flooding efficiency, that is, the better the reservoir physical property and stronger the seepage ability, the higher would be the CO₂ flooding efficiency.
- (iv) There is a positive correlation between oil flooding efficiency and injection pressure during CO₂ flooding in tight sandstone reservoirs, and the change in injection pressure has a significant impact on oil flooding efficiency. From near-miscible flooding to miscible flooding, the oil flooding efficiency increases rapidly. After entering the miscible flooding stage, with the increase in injection pressure, the increase in oil flooding efficiency decreases, which is flat compared with that in the previous stage.

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Notes

The authors declare no competing financial interest.

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